Explaining Efficiency-Accuracy Trade-offs in Sentiment Classification Models

Portfolio Report — Explainable AI Course, 2025–26

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Abstract

Deploying a sentiment classifier in practice requires balancing accuracy, inference efficiency, and interpretability – objectives that rarely align. This project investigates how these properties interact across model families, comparing three systems on the IMDb Large Movie Review benchmark: TF-IDF with Logistic Regression, TinyBERT quantized to INT8, and a cascade classifier that uses Logistic Regression probabilities for routing. We apply complementary explainability methods – global feature weight analysis for the sparse model and Local Interpretable Model-agnostic Explanations (LIME) for both model families. Integrated Gradients was planned for TinyBERT, but the repository retained only a quantized checkpoint, so gradient-based attribution was not run in the final setup. Evaluation is performed on a small clean sample derived from an original 100-row candidate file after removing exact train-set overlaps and duplicate texts, leaving 39 examples. On this clean sample, Logistic Regression reaches 97.4% accuracy (F1 0.973), TinyBERT reaches 92.3% (F1 0.919), and the cascade reaches 94.9% (F1 0.944). Logistic Regression runs at 0.28 ms per sample versus 9.88 ms for TinyBERT. The cascade with threshold $p \geq 0.80$ routes 56.4% of examples to Logistic Regression and reduces average latency to 5.19 ms per sample. The sparse model's explanations resemble a human-readable set of sentiment associations; LIME examples are presented only as illustrative local diagnostics. Given the small evaluation sample, the quantitative results are directional rather than statistically definitive.

1 Introduction

Sentiment classification underpins product review aggregation, customer feedback pipelines, social media monitoring, and recommendation systems. Despite the maturity of the field, deploying a sentiment classifier involves decisions that go beyond predictive performance: in many real-time pipelines, inference latency, memory footprint, and computational cost are hard constraints, and a slightly more accurate but significantly slower model may simply not be viable. Practitioners therefore consider a range of model families, from sparse linear classifiers built on bag-of-words features to compact neural models designed to approximate larger transformers at lower cost. Sparse linear models are attractive in latency-sensitive settings because they are fast, easy to deploy, and their decision logic can be inspected directly through learned feature weights. Unigrams and bigrams can capture local lexical patterns, but do not model broader context in the way transformers do.

What is less often discussed is how these architectural choices affect interpretability. A model whose predictions can be explained allows engineers to debug errors, audit for bias, and build user trust – value that goes beyond raw accuracy. Most XAI work studies individual model types in isolation; there is comparatively little research on how explanation patterns differ when the same task is approached by models from fundamentally different families, especially under efficiency constraints.

This project starts from a practical observation: efficiency-oriented practitioners do not choose a single model and explain it – they choose *between* model families, each with different interpretability properties. A TF-IDF linear model can be explained through its feature weights; a quantized transformer uses black-box approximation in this project because a matching fine-tuned float checkpoint was not retained; a cascade routes inputs dynamically, and the applicable explanation depends on which component handled the input. Our central question is: *how do accuracy, efficiency, and interpretability interact across different sentiment classification model families?* We investigate this through a controlled three-system comparison, characterising not just what each model gets right but how its explanations behave and where they break down.

2 Problem Definition

Task. We study binary sentiment classification over text reviews. Given a review $x \in \mathcal{X}$, the goal is to predict $y \in \{0, 1\}$ (0 = negative, 1 = positive). We use the IMDb Large Movie Review Dataset (Maas et al., 2011), a standard benchmark; no new data collection is performed.

Models. We define three classifier families $f_1, f_2, f_3 : \mathcal{X} \to \{0, 1\}$:

- f_1: **TF-IDF + Logistic Regression.** A TF-IDF map $\phi : \mathcal{X} \to \mathbb{R}^d$ feeds a logistic regression classifier with weights $\mathbf{w}$ and bias b: $\hat{y} = \mathbf{1}[\sigma(\mathbf{w}^\top \phi(x) + b) > 0.5]$.
- f_2: **TinyBERT INT8.** A compact transformer pre-trained, fine-tuned for sentiment, and post-training quantized to INT8 precision.
- f_3: **Cascade classifier.** Routes each input through f_1 first. If the maximum predicted class probability from Logistic Regression satisfies $c(x) = \max_k p(y{=}k \mid x) \geq \tau$, the cascade returns the sparse model's prediction; otherwise it falls back to f_2.

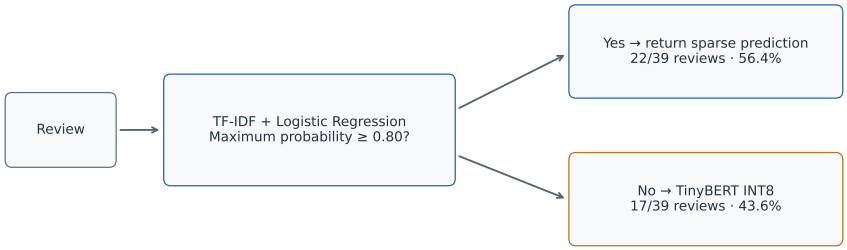


Figure 1: Cascade classifier routing logic. An input is first processed by Logistic Regression; if the predicted class probability meets the confidence threshold τ, its prediction is returned directly (fast path); otherwise TinyBERT INT8 is queried (fallback path).

Hardness. Sentiment classification is non-trivial whenever reviews use negation, sarcasm, or mixed sentiment. For linear models, the fundamental difficulty is representational: a vocabulary of size d admits $O(d^2)$ pairwise feature interactions, but a linear decision boundary in $\mathbb{R}^d$ assigns a fixed coefficient to each unigram or bigram feature. Bigrams encode limited adjacent-word information, but the model lacks broader contextual representations. Transformer models address this via self-attention, but their non-linear decision functions are harder to inspect directly. Here we use post-hoc local approximation as one available explanation route. The cascade further complicates interpretability: the applicable explanation method depends on which component handled each input, making the system a mixture model in both prediction and explanation.

Metrics. We evaluate accuracy, binary F1, precision, recall, and average per-sample inference time (ms) under identical CPU conditions.

Explainability. For f_1 we inspect the learned weight vector $\mathbf{w}$ globally. For local explanations we apply LIME (Ribeiro et al., 2016) to both f_1 and f_2: given input x, LIME fits a sparse linear model h over perturbed neighbours $\mathcal{N}(x)$, minimising

$$\sum_{z \in \mathcal{N}(x)} \pi_x(z) \, (f(z) - h(z))^2 + \Omega(h).$$

Integrated Gradients (Sundararajan et al., 2017) was planned for f_2 but was not run because the repository retained only a quantized checkpoint rather than a float model suitable for gradient attribution. LIME therefore serves as a model-agnostic fallback throughout.

3 Related Work

Post-hoc local explanations. Ribeiro et al. (2016) introduced LIME, which approximates a complex classifier locally via a sparse linear model fitted over perturbed inputs; it is model-agnostic but can vary between runs due to stochastic sampling. Lundberg and Lee (2017) proposed SHAP, grounding feature attributions in Shapley values from cooperative game theory. SHAP was not applied in this project; learned coefficients and feature-value-weighted contributions were used for the linear model.

Attribution for neural networks. Sundararajan et al. (2017) introduced Integrated Gradients, which integrates gradients along a path from a baseline to the input and satisfies an implementation invariance axiom; the method requires an appropriate differentiable model path. The retained quantized checkpoint did not provide a compatible path for our planned implementation.

Attention as explanation. Jain and Wallace (2019) showed that attention weights do not reliably correspond to other measures of feature importance; Wiegreffe and Pinter (2019) challenged this finding; and Serrano and Smith (2019) found that attended tokens are not always causally responsible for model outputs. Collectively, these results suggest attention is an unreliable explanation source.

Cross-family interpretability. Most prior work studies individual model types or explanation methods in isolation. This project addresses that gap by comparing sparse linear classifiers, a compact quantized transformer, and a cascade system, focusing on how architecture and efficiency constraints shape interpretability.

4 Methodology

Dataset and preprocessing. We use the IMDb Large Movie Review Dataset. For the sparse baseline, each review is represented as a TF-IDF weighted vector over unigrams and bigrams (n-gram range 1–2) using scikit-learn. The active pipeline lowercases text, removes English stop words, and keeps terms appearing in at least two training documents. TF-IDF weighting surfaces evaluative vocabulary by downweighting corpus-wide frequent terms.

Logistic Regression (f_1). Trained on the TF-IDF representation with L2 regularisation. The learned weight vector provides a direct global explanation: terms with large positive weights indicate positive sentiment and vice versa.

TinyBERT INT8 (f_2). We use TinyBERT (Jiao et al., 2020), a four-layer transformer obtained by knowledge distillation from BERT-base, making it substantially smaller and faster than its teacher while retaining much of its representational capability. TinyBERT was fine-tuned on the training set for sentiment classification and subsequently post-training dynamically quantized to INT8 for its linear layers. A memory comparison was not measured. Only the quantized fine-tuned checkpoint was retained, so the planned gradient-attribution path was not run.

Cascade classifier (f_3). Logistic Regression is queried first; the maximum predicted class probability is used as a routing score. If $c(x) \geq \tau$ the cascade returns f_1's prediction; otherwise it queries f_2. This keeps the routing criterion directly interpretable and tied to the actual classifier probabilities rather than an external heuristic.

Explainability pipeline. For f_1 we extract the top-ranked positive and negative feature weights globally. Coefficients are fixed for a retained model, but stability across retraining was not evaluated. For both f_1 and f_2 we apply LIME to a small illustrative set drawn from the first five rows of the clean evaluation sample. LIME describes local responses to word perturbations; its outputs can differ from global coefficients and can vary between runs. Integrated Gradients (planned for f_2) was not feasible in the final repository because only the quantized checkpoint was retained.

5 Experimental Setup

Evaluation sample. Final evaluation uses a small clean sample derived from an original 100-row candidate CSV by removing every review whose text appears exactly in the training file, then dropping duplicate texts inside the remaining sample. This leaves 39 labelled examples. The local training CSV contains 30,000 reviews, balanced across labels. Exactly 61 candidate rows overlap its raw review text; no additional duplicates remain after exclusion. This custom project partition is not the official IMDb 25,000-review test split. The split-creation logic is saved in a short script, and file hashes record the input identities. Because the resulting sample is small, all quantitative differences should be treated as indicative rather than statistically reliable.

Evaluation metrics. For each model we report accuracy, binary F1, precision, recall, and average wall-clock inference time per sample (ms) over the full evaluation sample. Precision and recall are reported separately because the two main model families exhibit distinct error profiles.

Hardware. All models were evaluated on CPU without GPU acceleration, consistent with the project's focus on resource-constrained deployment. No cross-validation was performed, as the primary goal is comparative explanation analysis rather than exhaustive model optimisation.

Timing protocol. To keep latency comparisons fair, all three models are timed in the same per-sample prediction loop on CPU. Each evaluation is repeated three times and the average wall-clock time is reported.

Cascade threshold. We use a single probability threshold $\tau = 0.80$ for the final cascade. No threshold sweep is claimed in the final analysis.

Explainability evaluation. The analysis is primarily qualitative. We examine global feature weights for the linear models and LIME explanations on the first five rows of the clean evaluation sample. A formal stability analysis or large-scale quantitative comparison is not conducted; the evaluation sample is too small for statistically meaningful conclusions of that kind.

6 Results

6.1 Quantitative Performance

Table 1 summarises classification performance and inference time on the clean evaluation sample.

Table 1: Classification performance and inference time on the clean evaluation sample ($n = 39$). Speedup is relative to TinyBERT INT8.

Model	Acc.	F1	Prec.	Recall	ms/samp.	Speedup
TF-IDF + LR	0.974	0.973	1.000	0.947	0.28	35.3×
TinyBERT INT8	0.923	0.919	0.944	0.895	9.88	1×
Cascade (τ=0.80)	0.949	0.944	1.000	0.895	5.19	1.91×

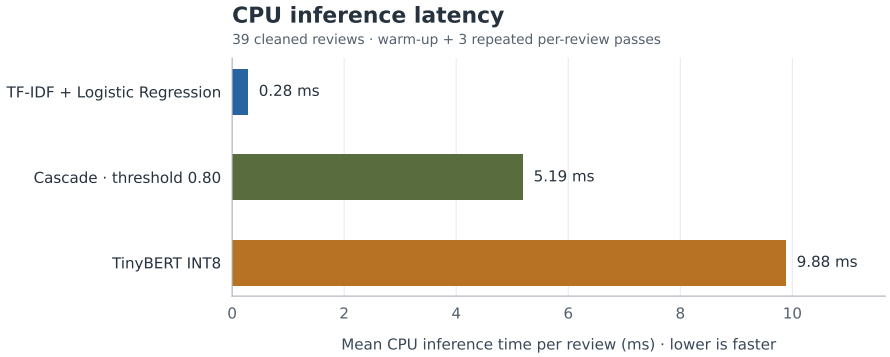


Figure 2: Average inference latency per sample on CPU. Logistic Regression is approximately 35× faster than TinyBERT INT8 on the final clean evaluation sample (n=39, τ=0.80). The figure is generated directly from the saved clean evaluation JSON used by Table 1.

On the clean evaluation sample, TF-IDF + Logistic Regression achieves the best predictive result: 38 correct predictions out of 39, with accuracy 97.4%, precision 1.00, recall 0.947, and binary F1 0.973. TinyBERT INT8 reaches 92.3% accuracy and F1 0.919, while the cascade sits between the two at 94.9% accuracy and F1 0.944. Because the clean sample is so small, these differences should still be read cautiously; they are based on only one or two reviews.

The clearest quantitative difference is still efficiency. Logistic Regression runs at 0.28 ms per sample, compared with 9.88 ms for TinyBERT INT8 a gap of roughly 35× under the shared timing setup. The cascade at $\tau = 0.80$ averages 5.19 ms, reflecting the fact that it can keep part of the traffic on the sparse model while still falling back to TinyBERT when Logistic Regression is uncertain.

6.2 Cascade Routing Behaviour

At $\tau = 0.80$, the cascade routed 22 of the 39 clean evaluation examples (56.4%) to Logistic Regression and 17 (43.6%) to TinyBERT. On this sample it improves on TinyBERT's latency while also improving on TinyBERT's accuracy, but it does not match the sparse model's best result. This does not prove that the cascade is universally superior; on such a small sample it simply shows that a basic probability gate can recover some of the sparse model's speed while keeping predictive performance in the same range.

Table 2: Observed cascade routing at the single threshold used in the final analysis.

τ	LR routed	BERT routed	Accuracy	ms/samp.
0.80	**56.4%**	**43.6%**	**94.9%**	**5.19**

The routing interpretation is straightforward because the gate is based on Logistic Regression probabilities rather than a hand-designed confidence score. That makes the final cascade section easier to defend, even though the result still rests on a very small evaluation sample.

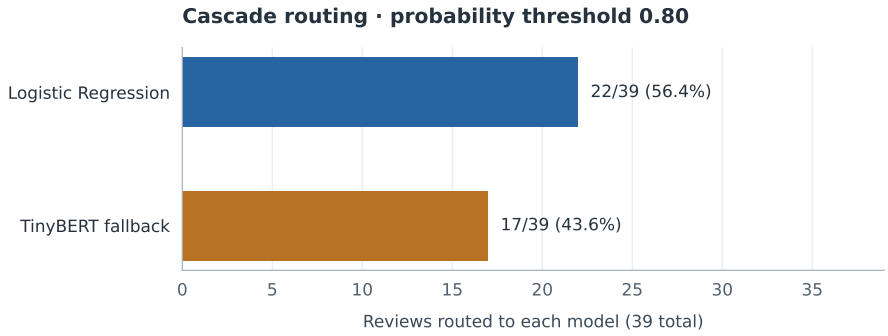


Figure 3: Cascade routing on the 39-review clean evaluation sample at τ=0.80: 22 reviews (56.4%) used Logistic Regression, and 17 reviews (43.6%) used TinyBERT.

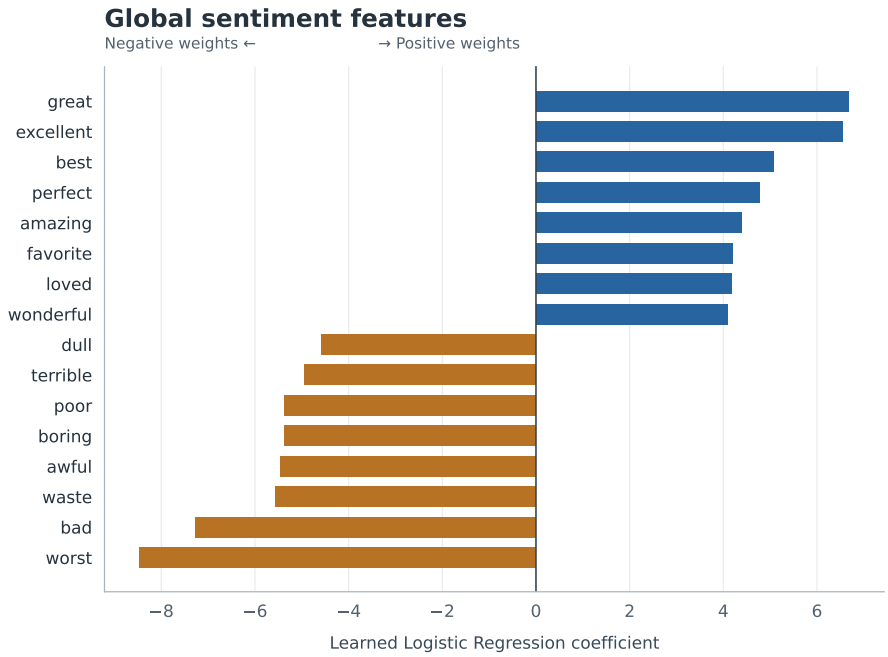


Figure 4: Top positive and negative feature weights learned by TF-IDF + Logistic Regression. Blue bars indicate terms associated with positive sentiment; orange bars indicate terms associated with negative sentiment.

6.3 Global Explanation Analysis

Inspecting the top-ranked feature weights of the Logistic Regression model reveals a pattern closely resembling a manually constructed sentiment lexicon. The most negatively weighted terms include *worst*, *bad*, *waste*, *boring*, and *awful*; the most positively weighted include *excellent*, *great*, *amazing*, and *brilliant*. This is consistent with what one would expect from a model trained on a sentiment benchmark with clear evaluative language. These coefficients expose learned lexical associations. Bigrams provide some adjacent-word information, while longer-range interactions are absent. Some weights are less interpretable – words such as *today* and *minutes* appear with non-trivial weights, likely reflecting dataset artefacts (co-occurrence with phrases like *waste of two minutes*) rather than genuine sentiment signal. No causal error study of these features was performed. They illustrate why a learned coefficient should be interpreted as a model association, rather than as an independently verified sentiment rule.

6.4 Local Explanation Analysis

LIME explanations were computed for the first five rows of the clean evaluation sample. All five are retained in each model's saved JSON output; the figure shows the first two. The examples are illustrative and were not selected as a representative evaluation of explanation quality.

Example 1. Both models predict negative sentiment. The token *disappointed* has negative LIME weight for both

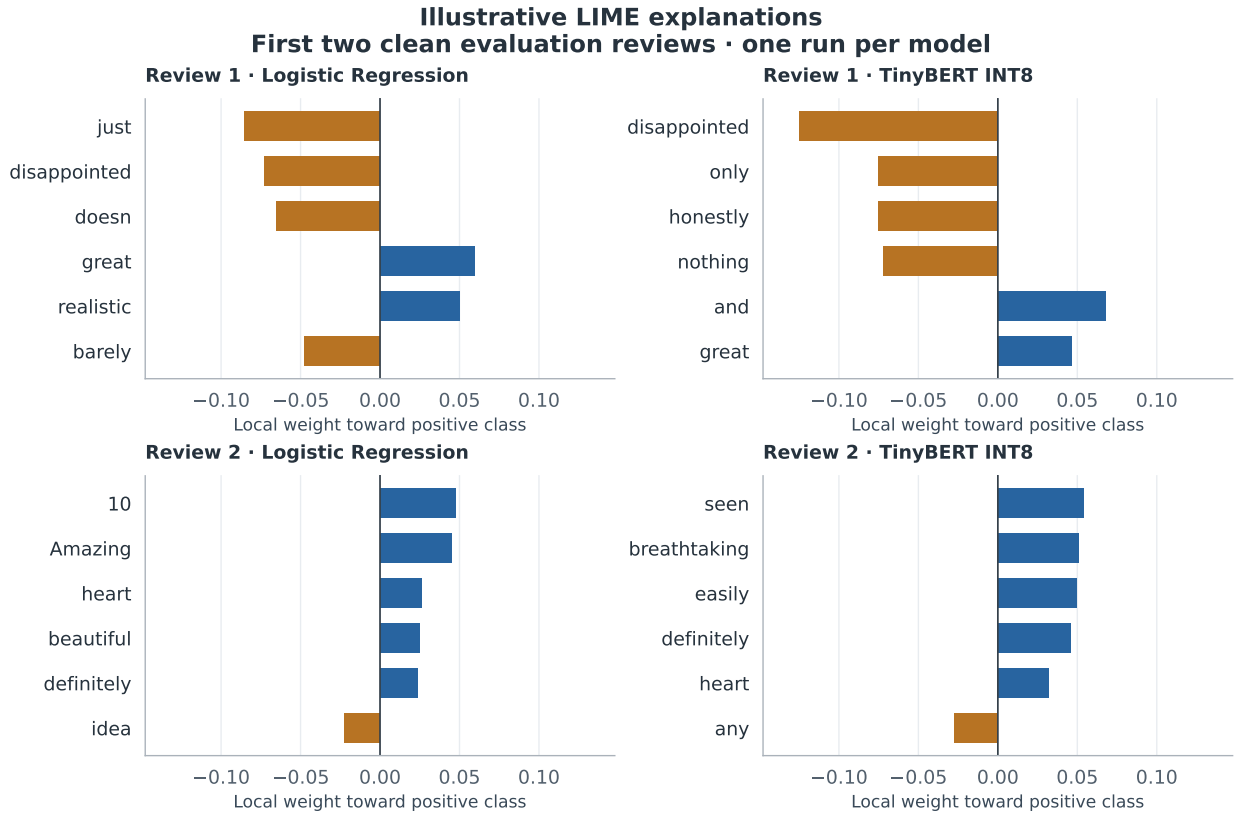


Figure 5: LIME local explanations for the first two clean evaluation reviews. Both columns use the same horizontal scale. Blue weights support the positive class and orange weights oppose it. These saved runs illustrate local perturbation responses; they do not establish explanation stability or a causal account of model reasoning.

models, while *great* has positive weight. Their other highly ranked terms differ: Logistic Regression emphasizes *just*, while TinyBERT includes *only* and *honestly* among its negative weights.

Example 2. Both models predict positive sentiment. Logistic Regression assigns positive local weight to *Amazing* and *beautiful*; TinyBERT assigns positive weight to *breathtaking* and *easily*. Both give positive weight to *heart* and *definitely*.

These are descriptions of saved surrogate-model coefficients. Differences in word rankings do not by themselves prove that TinyBERT resolves context more faithfully. LIME is stochastic, and neither repeated-seed stability nor explanation faithfulness was evaluated.

6.5 Failure Cases and Limitations of Explanation Methods

The saved aggregate metrics show one error for Logistic Regression, three for TinyBERT, and two for the cascade. They do not identify which review belongs to each cross-model error category. We therefore do not attribute a specific sarcasm or negation failure to these aggregate results.

A future error analysis should retain per-review predictions and compare negation, mixed sentiment, long reviews, and ambiguous phrasing on a larger independent sample. It should distinguish actual observed errors from examples constructed to illustrate an architectural limitation.

Limitations of explanation methods. The most significant constraint is that Integrated Gradients was not run for TinyBERT because the repository retained only a quantized checkpoint. This should not be phrased as a general claim that INT8 models can never support gradient-based XAI; the narrower point is that saving only the quantized artifact removed that option in this project. Saving a float32 checkpoint before quantization would preserve the ability to run gradient-based attribution later.

7 Conclusions

This project investigated how accuracy, efficiency, and interpretability interact across three sentiment classification systems. Evaluation was conducted on a small clean sample of 39 examples created by removing exact train-set overlap from an original 100-row candidate file; stronger conclusions would require a substantially larger evaluation.

On performance, Logistic Regression gives the strongest result on this clean sample and remains dramatically faster on CPU. Because the sample contains only 39 rows, that should still be framed cautiously; the most reliable conclusion is about the large efficiency gap, not about a decisive accuracy ordering between model families.

On interpretability, the retained sparse model offers inspectable global feature weights that reveal lexical sentiment associations. TinyBERT required model-agnostic local explanations, and those LIME plots are useful only as qualitative diagnostics. The cascade with $\tau = 0.80$ shows that a simple probability gate can reduce latency relative to TinyBERT alone while preserving most of the sparse model's performance on this small sample.

The unavailability of Integrated Gradients reflects a practical repository constraint: retaining only a quantized checkpoint prevented gradient-based XAI analysis in the final submission. Saving a float32 checkpoint before quantization would preserve that option, with an additional model-storage requirement.

Future work. Evaluation on a substantially larger test set would enable reliable statistical comparisons and explanation stability analysis. Retaining a float32 TinyBERT checkpoint before quantization would allow Integrated Gradients to be applied, enabling a principled comparison between gradient-based and perturbation-based explanations.

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